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Collector and cylindrical secondary battery including the same

Abstract

A cylindrical secondary battery includes an electrode assembly provided with an electrode tab; a battery can configured to accommodate the electrode assembly; a terminal configured to pass through the battery can and insulated from the battery can; and a collector configured to electrically connect the electrode tab to the terminal. The collector may include an edge part; a terminal coupling part disposed inside the edge part and coupled to the terminal; a bridge configured to connect the edge part to the terminal coupling part; and a tab coupling part extending inward from the edge part, spaced apart from the bridge and the terminal coupling part, and coupled to the electrode tab. An inner end of the tab coupling part may have a width that gradually decreases as it approaches the terminal coupling part.

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATION**

(1) The present application is the National Phase of PCT International Application No. PCT/KR2023/001142, filed on Jan. 25, 2023, which claims priority under 35 U.S. C. 119(a) to Patent Application No. 10-2022-0010936, filed in Republic of Korea on Jan. 25, 2022, all of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) The present disclosure relates to a collector and a cylindrical secondary battery including the same.

BACKGROUND ART

(3) A cylindrical secondary battery according to a related art generally has a structure in which a tab connecting a jelly-roll to an external terminal is welded to a foil of the jelly-roll. The cylindrical secondary battery having such a structure has a limited current path, and self-resistance of the jelly-roll is inevitably very high.

(4) Thus, since a method for reducing resistance by increasing a number of tabs connecting the jelly-roll to an external terminal has been tried, there is a limit to reduce resistance to a desired level and sufficiently secure a path of current only by increasing a number of tabs as described above.

(5) Thus, it is necessary to develop a new jelly-roll structure and a structure of a collector suitable for the jelly-roll structure in order to reduce the self-resistance of the jelly-roll. Particularly, the jelly-roll having the new structure and the application of the collector are more necessary for devices that require a battery pack having high output/high capacity, such as, for example, an electric vehicle.

(6) In addition, in the case of the battery pack applied to the electric vehicle and the like, it is inevitably exposed to a lot of vibration and impact when considering the use environment. Therefore, there is a need to develop a cylindrical secondary battery having a structure in which there is little risk of damage to a welded portion even when vibration and an external impact are applied, and a collector structure applied to the cylindrical secondary battery.

SUMMARY OF THE DISCLOSURE**Technical Problem**

(7) One object to be implemented by the present disclosure is to provide a collector having a

structure suitable for an electrode assembly having a low resistance structure, and a cylindrical secondary battery including the same.

(8) Another object to be implemented by the present disclosure is to provide a collector having a structure that is capable of reducing possibility of damage to a welded portion with an electrode assembly and/or a welded portion with a battery can even when vibration and an impact are applied, and a cylindrical secondary battery including the same.

Technical Solution

(9) A cylindrical secondary battery according to an aspect of the present disclosure may include: an electrode assembly having an electrode tab; a battery can configured to accommodate the electrode assembly therein; a terminal configured to pass through the battery can while being from the battery can; and a collector configured to electrically connect the electrode tab to the terminal. The collector may include: an edge part; a terminal coupling part located inside the edge part, the terminal coupling part being coupled to the terminal; a bridge connecting the edge part to the terminal coupling part; and a tab coupling part extending inward from the edge part, the tab coupling part being spaced apart from the bridge and the terminal coupling part, the tab coupling part being coupled to the electrode tab. An inner end of the tab coupling part may have a width that gradually decreases as it approaches the terminal coupling part.

(10) The bridge and the tab coupling part may be provided in plurality and may be alternately disposed in a circumferential direction of the collector.

(11) The outer end of the bridge may have a width that gradually increases as it approaches the edge part.

(12) The bridge includes a narrow part located between opposite ends of the bridge.

(13) The narrow part may have a width corresponding to 70% to 80% of a width of the bridge at one of the ends of the bridge.

(14) The narrow part may be disposed closer to an outer end of the opposite ends of the bridge.

(15) A distance from a center of the collector to the narrow part may be 0.4 times to 0.9 times a radius of the electrode assembly.

(16) At least a portion of the electrode tab may be bent toward a central axis of the electrode assembly, and the narrow part may be opposite the bent portion of the electrode tab in an axial direction of the electrode assembly.

(17) The narrow part may be defined by recessed grooves at opposite edges of the bridge or a hole extending through the bridge.

(18) The narrow part may have a width less than that of a width of the bridge between the edges of the bridge.

(19) The bridge may be provided with a cover member configured to surround the bridge and may be made of an insulating material having thermal conductivity less than that of the bridge.

(20) The cover member may be made of a polyimide (PI) material.

(21) The edge part may have a width less than a width of the tab coupling part.

(22) The collector may have a radius corresponding to 33% to 102% of a radius of the electrode assembly.

(23) The collector may have a radius corresponding to 33% to 98.5% of an inner radius of the battery can.

(24) The terminal may include a body part coupled to the terminal coupling part and the terminal coupling part may have a radius corresponding to 40% to 320% of a radius of the body part.

(25) At least one opening may be defined between the edge part and the terminal coupling part, and a ratio of a surface area of the at least one opening to a total surface area of the collector may be 40% to 99%.

(26) A collector according to an aspect of the present disclosure may electrically connect an electrode assembly to a terminal passing through a battery can in which the electrode assembly is accommodated. The collector may include: an edge part; a terminal coupling part located inside the

edge part, the terminal coupling part may be coupled to the terminal; a bridge connecting the edge part to the terminal coupling part; and a tab coupling part extending inward from the edge part, the tab coupling part being spaced apart from the bridge and the terminal coupling part, the tab coupling part being coupled to an electrode tab provided in the electrode assembly. An inner end of the tab coupling part may have a width that gradually decreases as it approaches the terminal coupling part.

(27) The bridge and the tab coupling part may be provided in plurality, and the bridges and the tab coupling parts are alternately disposed in a circumferential direction of the collector.

(28) An outer end of the bridge may have a width that gradually increases as it approaches the edge part.

(29) The bridge may include a narrow part located between opposite ends of the bridge.

(30) The narrow part may be defined by recessed grooves at opposite edges of the bridge or as a hole extending through the bridge.

(31) The cylindrical secondary battery may be provided with a cover member configured to surround the bridge and may be made of an insulating material having thermal conductivity less than that of the bridge.

(32) A cylindrical secondary battery according to an aspect of the present disclosure may include: an electrode assembly having a first electrode tab and a second electrode tab; a battery can configured to accommodate the electrode assembly therein; a terminal configured to pass through the battery can while being spaced from the battery can; a first collector configured to electrically connect the first electrode tab to the battery can; and a second collector configured to electrically connect the second electrode tab to the terminal.

(33) A difference between a coupling area between the first electrode tab and the first collector and a coupling area between the second electrode tab and the second collector may be three times or less.

(34) The battery can may include a beading part on a circumference of the battery can, the beading part extending inward between the cap plate and the electrode assembly, and the first collector may include a can coupling part coupled to the beading part.

(35) The second collector may not overlap the beading part in a height direction of the electrode assembly.

(36) The second collector may overlap the beading part in a height direction of the electrode assembly, and a width of an area of the beading part, which overlaps the second collector, may be 80% or less of a width of the beading part.

Advantageous Effects

(37) According to an aspect of the present disclosure, when the electrode assembly and the battery can are electrically connected to each other, the resistance may be significantly reduced.

(38) In addition, even when vibration and impact are applied while the secondary battery is used, the possibility of damage to the welded portion between the collector and the electrode assembly and/or the welded portion between the collector and the battery can may be significantly reduced.

(39) In addition, the effects that are obvious to those skilled in the art may be predicted from the configurations according to the various aspects of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings attached in this specification illustrate an aspect of the present disclosure and function to make further understood the technical spirit of the present disclosure along with the detailed description of the disclosure, and thus, the present disclosure should not be construed as being limited to only the drawings.

(2) FIG. 1 is a perspective view of a cylindrical secondary battery according to an aspect of the present disclosure.

(3) FIG. 2 is a cutaway cross-sectional view of the cylindrical secondary battery according to an aspect of the present disclosure.

(4) FIG. 3 is an enlarged cross-sectional view of a first collector and surroundings of the first collector, which are illustrated in FIG. 2.

(5) FIG. 4 is a perspective view of the first collector according to an aspect of the present disclosure.

(6) FIG. 5 is an enlarged cross-sectional view of a second collector and surroundings of the second collector, which are illustrated in FIG. 2.

(7) FIG. 6 is a plan view of the second collector according to an aspect of the present disclosure.

(8) FIG. 7 is an enlarged view illustrating a narrow part illustrated in FIG. 6.

(9) FIG. 8 is a view illustrating a modified example of the narrow part illustrated in FIG. 6.

(10) FIG. 9 is a view illustrating a state in which a cover member surrounds the narrow part illustrated in FIG. 6.

(11) FIG. 10 is a plan view of a second collector according to another aspect of the present disclosure.

MODE FOR CARRYING OUT THE DISCLOSURE

(12) Hereinafter, various aspects of the present disclosure will be described in detail with reference to the accompanying drawings so that those of ordinary skill in the art can easily carry out the present disclosure. However, the present disclosure may be implemented in several different forms and is not limited or restricted by the following examples.

(13) In order to clearly explain the present disclosure, detailed descriptions of portions that are irrelevant to the description or known related technologies that may unnecessarily obscure the gist of the present disclosure have been omitted, and in the present specification, reference symbols are added to components in each drawing. In this case, the same or similar reference numerals are assigned to the same or similar elements throughout the specification.

(14) Also, terms or words used in this specification and claims should not be restrictively interpreted as ordinary meanings or dictionary-based meanings, but should be interpreted as meanings and concepts conforming to the scope of the present disclosure on the basis of the principle that an inventor can properly define the concept of a term to describe and explain his or her disclosure in the best ways.

(15) FIG. 1 is a perspective view of a cylindrical secondary battery according to an aspect of the present disclosure, and FIG. 2 is a cutaway cross-sectional view of the cylindrical secondary battery according to an aspect of the present disclosure.

(16) A cylindrical **1** secondary battery **1** (hereinafter, referred to as 'secondary battery') according to an aspect of the present disclosure may include an electrode assembly **10** and a battery can **20** accommodating the electrode assembly **10**.

(17) The electrode assembly **10** may include a positive electrode, a negative electrode, and a separator interposed between the positive electrode and the negative electrode. In addition, the electrode assembly **10** may be provided with a first electrode tab **11** connected to one of the positive electrode and the negative electrode, and a second electrode tab **12** connected to the other of the positive electrode and the negative electrode.

(18) In more detail, the electrode assembly **10** may be a jelly-roll type. The electrode assembly **10** may be manufactured by winding a stack in which the positive electrode, the separator, the negative electrode, and the separator are sequentially stacked. The electrode assembly **10** may preferably have a height-to-diameter ratio of 1 or more, but is not limited thereto. A hollow C defined to be elongated in a height direction may be defined in a center of the electrode assembly **10**. In order to insulate the electrode assembly **10** from an inner circumference of the battery can **20**, the separator may be disposed at the outermost side of the electrode assembly **10**.

(19) The first electrode tab **11** and the second electrode tab **12** may be provided on opposite ends of the electrode assembly **10** in the height direction, respectively. In more detail, a non-coating portion of any one of the positive electrode and the negative electrode may be disposed on one end of the electrode assembly **10** to function as a first electrode tab **11**. The non-coating portion of the other of the positive electrode and the negative electrode may be disposed on the other end of the electrode assembly **10** to function as a second electrode tab **12**.

(20) Each of the electrode tabs **11** and **12** may extend along a winding direction of the electrode assembly **10**.

(21) Alternatively, each of the electrode tabs **11** and **12** may include a plurality of foil-flags disposed along the winding direction of the electrode assembly **10**.

(22) A first collector **50** to be described below may be coupled to the first electrode tab **11**, and a second collector **60** to be described below may be coupled to the second electrode tab **12**. The electrode assembly **10** may be accommodated in the battery can **20** in a state in which the collectors **50** and **60** are coupled to the electrode tabs **11** and **12**, respectively.

(23) For example, the first electrode tab **11** may be a non-coating portion of the negative electrode, and the second electrode tab **12** may be a non-coating portion of the positive electrode. In this case, the first collector **50** may be referred to as a negative electrode collector, and the second collector **60** may be referred to as a positive electrode collector.

(24) The battery can **20** may have a substantially cylindrical shape. One surface **20a** may be provided at one side of the battery can **20** in the height direction, and an opening may be defined at the other side.

(25) The electrode assembly **10** may be accommodated in the battery can **20** together with an electrolyte through the opening.

(26) The battery can **20** may have a conductive material such as a metal. The battery can **20** may be electrically connected to the first electrode tab **11** through the first collector **50** to be described below. The battery can **20** may have the same polarity as the first electrode tab **11**.

(27) A beading part **21** and a crimping part **22** may be provided on the battery can **20**. The beading part **21** and the crimping part **22** may be disposed adjacent to the opening of the battery can **20**.

(28) The beading part **21** may be provided by press-fitting a circumference of the battery can **20** inward in a radial direction. An inner diameter of the beading part **21** may be less than an inner diameter of another portion of the battery can **20**.

(29) The beading part **21** may be provided in a state in which the electrode assembly **10** is accommodated in the battery can **20**. The electrode assembly **10** may be restricted between the one surface **20a** of the battery can **20** and the beading part **21** in the height direction of the battery can **20**. The beading part **21** may prevent the electrode assembly **10** from being separated from the opening of the battery can **20**. In addition, the beading part **21** may be coupled to the first collector **50**. For example, the first collector **50** may be welded to the beading part **21**. Thus, the first collector **50** may electrically connect the first electrode tab **11** to the battery can **20**.

(30) The crimping part **22** may be disposed adjacent to the beading part **21**. The crimping part **22** may be provided by bending an end of a circumferential surface of the battery can **20** inward in a radial direction. The crimping part **22** may be formed after the first collector **50** is coupled to the beading part **21**. The crimping part **22** may restrict a cap plate **30**, which will be described below, together with the beading part **21**.

(31) The secondary battery **1** may include the cap plate **30** covering the opening of the battery can **20**.

(32) The cap plate **30** may have a substantially circular plate shape. The cap plate **30**, together with the battery can **20**, may define an outer appearance of the secondary battery **1**. An edge portion of the cap plate **30** may be restricted between the beading part **21** and the crimping part **22** of the battery can **20**.

(33) The cap plate **30** may have a material having high rigidity such as a metal. The cap plate **30**

may be insulated from the battery can **20** and may not have a polarity. In more detail, a sealing part **72** of a spacer **70** may be interposed between the cap plate **30** and the battery can **20**. This will be described in detail below.

(34) The secondary battery **1** may include a terminal **40** fixed by passing through the one surface **20a** of the battery can **20**.

(35) The terminal **40** may pass through the one surface **20a** of the battery can **20**, more specifically, a center of the one surface **20a**. A portion of the terminal **40** may be exposed to the outside of the battery can **20**, and another portion of the terminal **40** may be disposed inside the battery can **20**. For example, the terminal **40** may be fixed to the one surface **20a** of the battery can **20** by riveting.

(36) The terminal **40** may have a conductive material such as a metal.

(37) The terminal **40** may be coupled to the second collector **60** to be described below. For example, the second collector **60** may be welded to the terminal **40**. Thus, the second collector **60** may electrically connect the second electrode tab **12** to the terminal **40**.

(38) The terminal **40** may be insulated from the battery can **20**. In more detail, an insulating gasket **G** may be interposed between the terminal **40** and the battery can **20**. However, it is not limited thereto, and an insulating coating layer may be disposed on a portion of the terminal **40**, or a method for structurally fixing the terminal **40** may be applied while spacing the terminal **40** and the battery can **20** apart from each other.

(39) The insulating gasket **G** may be deformed during the riveting of the terminal **40** and then be bent toward an inner surface of the one surface **20a** of the battery can **20**. The insulating gasket **G** may seal a gap between the terminal **40** and the battery can **20**.

(40) That is, the battery can **20** and the terminal **40** may have polarities opposite to each other due to the insulating gasket **G**. In more detail, the battery can **20** may have the same polarity as the first electrode tab **11**, and the terminal **40** may have the same polarity as the second electrode tab **12**. Thus, the one surface **20a** of the battery can **20** may function as a first terminal having a first polarity, and the terminal **40** may function as a second terminal having a second polarity. Since the one surface **20a** of the battery can **20** and the terminal **40** are disposed adjacent to each other, an inner space required for installing a bus bar in the battery module including the plurality of secondary batteries **1** may be reduced, and thus, energy density of the battery module may increase.

(41) The secondary battery **1** may include the first collector **50** electrically connecting the first electrode tab **11** to the battery can **20**. For example, the first electrode tab **11** may be connected to the negative electrode of the electrode assembly **10**, and the first collector **50** may be referred to as a negative electrode collector.

(42) The secondary battery **1** may include the second collector **60** electrically connecting the second electrode tab **12** to the terminal **40**. For example, the second electrode tab **12** may be connected to the positive electrode of the electrode assembly **10**, and the second collector **60** may be referred to as a positive collector.

(43) The first collector **50** and the second collector **60** may be disposed within the battery can **20**.

(44) The first collector **50** may be disposed between the electrode assembly **10** and the cap plate **30** in the height direction of the battery can **20**. The first collector **50** may be coupled to the first electrode tab **11**. In more detail, the first collector **50** may be welded to the first electrode tab **11**. In addition, the first collector **50** may be welded to the battery can **20**, more specifically, to the beading part **21**.

(45) The second collector **60** may be disposed between the electrode assembly **10** and the one surface **20a** of the battery can **20** in the height direction of the battery can **20**. The second collector **60** may be coupled to the second electrode tab **12**. In more detail, the second collector **60** may be welded to the second electrode tab **12**. In addition, the second collector **60** may be welded to the terminal **40**.

(46) The secondary battery **1** may include a spacer **70** disposed between the cap plate **30** and the electrode assembly **10**. In more detail, the spacer **70** may be disposed between the cap plate **30** and

the first collector **50**.

(47) A height of the spacer **70** may correspond to a distance between the cap plate **30** and the electrode assembly **10**. The spacer **70** may prevent the electrode assembly **10** from moving or shaking within the battery can **20**.

(48) The secondary battery **1** may include an insulator **80** disposed between the electrode assembly **10** and the battery can **20**. In more detail, the insulator **80** may be disposed between the second collector **60** and the battery can **20**.

(49) The insulator **80** may insulate the second collector **60** and the electrode assembly **10** from the battery can **20**. The insulator **80** may have a resin material having insulating properties.

(50) A hole through which the terminal **40** passes may be defined in a center of the insulator **80**. The terminal **40** may be coupled to the second collector **60** through the hole.

(51) The insulator **80** may have a substantially cap shape. In this case, a portion of the insulator **80** may be disposed between the one surface **20a** of the battery can **20** and the second collector **60**, and another portion of the insulator **80** may be disposed between an outer circumference of the second collector **60** and an inner circumference of the battery can **20**.

(52) However, it is not limited thereto, and the insulator **80** may have a plate shape disposed between the one surface **20a** of the battery can **20** and the second collector **60**. In this case, an insulating tape that is separate from the insulator **80** may be interposed between the outer circumference of the second collector **60** and the inner circumference of the battery can **20**.

(53) A height of the insulator **80** may correspond to a vertical distance between the one surface **20a** of the battery can **20** and the electrode assembly **10**. The insulator **80** together with the spacer **70** may prevent the electrode assembly **10** from moving or shaking within the battery can **20**.

(54) FIG. 3 is an enlarged cross-sectional view of the first collector and surroundings of the first collector, which are illustrated in FIG. 2, and FIG. 4 is a perspective view of the first collector according to an n aspect of the present disclosure.

(55) As described above, the first collector **50** may be coupled to the first electrode tab **11**. In more detail, at least a portion of the first electrode tab **11** may be bent toward a hollow C of the electrode assembly **10**, and the first collector **50** may be coupled to the bent portion of the first electrode tab **11**. Bent portions of the plurality of first electrode tabs **11** adjacent to each other may overlap each other.

(56) As a result, a height of the first electrode tab **11** may be reduced, and energy density of the secondary battery **1** may be improved. In addition, since a coupling area between the first electrode tab **11** and the first collector **50** increases, coupling force between the first electrode tab **11** and the first collector **50** may be improved to reduce resistance.

(57) In more detail, the first collector **50** may include a center part **51**, a tab coupling part **52** extending outward from the center part **51** and coupled to the first electrode tab **11**, a can coupling part **53** coupled to the battery can **20** and spaced apart from the tab coupling part **52**, and a bridge **54** connecting the can coupling part **53** to the center part **51**.

(58) The center part **51** may be disposed at a center of one side (e.g., lower side) of the electrode assembly **10**. The center part **51** may have an approximately circular shape.

(59) A center hole **51a** may be defined in the center part **51**. The center hole **51a** may face the hollow C of the electrode assembly **10**. The center hole **51a** together with the hollow C may function as a passage for insertion of a welding rod for the coupling between the terminal **40** and the second collector **60** or laser irradiation. In addition, the center hole **51a** may function as a passage through which an electrolyte is smoothly impregnated into the electrode assembly **10** when the electrolyte is injected.

(60) The tab coupling part **52** may be expanded radially outward from the center part **51**. The tab coupling part **52** may be coupled to the first electrode tab **11**. The tab coupling part **52** may be provided in plurality, and the plurality of tab coupling parts **52** may be radially expanded from the center part **51**.

(61) A width of the tab coupling part **52** may increase as a distance from the center part **51** increases. Thus, the tab coupling part **52** may widely cover the bent portion of the first electrode tab **11** and may prevent the bent portion of the first electrode tab **11** from being lifted.

(62) Central axes of each of the electrode assembly **10**, the battery can **20**, and the first collector **50** may substantially coincide with each other. A distance **R1** from the central axis of the electrode assembly **10** to an outer edge of the tab coupling part **52** may be less than a radius **Ra** of the electrode assembly **10** and also be equal to or less than an inner radius **Rb** of the beading part **21** in the radial direction of the battery can **20**.

(63) In more detail, the outer edge of the tab coupling part **52** may have an arc shape having a first curvature radius **R1**. A center of curvature of the outer edge may coincide with the center of the first collector **50**.

(64) The first curvature radius **R1** may be less than the radius **Ra** of the electrode assembly **10**. In addition, the first curvature radius **R1** may be less than the inner radius **Rb** of the beading part **21** disposed on the battery can **20**.

(65) The tab coupling part **52** may not overlap the beading part **21** in the height direction of the battery can **20**. That is, the tab coupling part **52** may not be disposed between the electrode assembly **10** and the beading part **21**. Thus, it is possible to prevent the tab coupling part **52** from being damaged by being jammed between the electrode assembly **10** and the beading part **21** due to a sizing process. The sizing refers to a compression process of reducing a height of the beading part **21** to reduce a height of the secondary battery **1** after the beading part **21** and the crimping part **22** are formed on the battery can **20** in which the electrode assembly **10** is accommodated.

(66) The distance **R1** from the central axis of the electrode assembly **10** to the outer edge of the tab coupling part **52** may be equal to or greater than $\frac{2}{3}$ of the radius **Ra** of the electrode assembly **10**. In more detail, the first curvature radius **R1** may be $\frac{2}{3}$ or more of the radius **Ra** of the electrode assembly **10**. Thus, a sufficiently wide contact area of the tab coupling part **52** with respect to the electrode tab **11** of the electrode assembly **10** may be secured.

(67) Both edges of the tab coupling part **52** may be disposed along the radial direction of the first collector **50**. A virtual line extending along both the edges may pass through the center of the first collector **50**.

(68) An injection hole **52d** may be provided in the tab coupling part **52**. The impregnation of the electrolyte into the electrode assembly **10** may be improved by the injection hole **52d**.

(69) The width of the injection hole **52d** may increase as a distance from the center part **51** increases. As a result, the electrolyte may be impregnated more uniformly with respect to the electrode assembly **10** in which a winding length per unit angle becomes longer toward the outside of the radius.

(70) The can coupling part **53** may be coupled to the battery can **20**, more specifically to the beading part **21**. The can coupling part **53** may be coupled to a surface disposed relatively far from the electrode assembly **10** on the beading part **21**.

(71) The can coupling part **53** may have a predetermined height difference with respect to the center part **51** and the tab coupling part **52**. The height difference may mean a distance between the tab coupling part **52** and the can coupling part **53** with respect to the height direction of the battery can **20**. The height difference may correspond to the height of the beading part **21**. That is, the distance between the tab coupling part **52** and the can coupling part **53** in the height direction of the battery can **20** may be the same as or similar to the height of the beading part **21**.

(72) The can coupling part **53** may be pressed by a sealing part **72** to be described below. In more detail, the can coupling part **53** may be interposed and fixed between the sealing part **72** and the beading part **21**.

(73) The can coupling part **53** may be spaced apart from the tab coupling part **52**. In more detail, each of the can coupling part **53** and the tab coupling part **52** may be only connected to the center part **51**, and the can coupling part **53** and the tab coupling part **52** may not be directly connected to

each other. As a result, when an impact or vibration is applied to the secondary battery **1**, stress applied to the coupled portion between the tab coupling part **52** and the first electrode tab **11** and the coupled portion between the can coupling part **53** and the battery can **20** are dispersed, and the possibility of occurrence of damage to the first collector **50** may be minimized.

(74) The can coupling part **53** may be provided in plurality. The can coupling part **53** and the tab coupling part **52** may be alternately disposed in a circumferential direction of the first collector **50**.

(75) The can coupling part **53** may extend along the inner circumference of the battery can **20**. That is, the can coupling part **53** may extend in the circumferential direction of the beading part **21**.

Since a contact area between the can coupling part **53** and the beading part **21** increases, the can coupling part **53** and the beading part **21** may be stably coupled to each other, and the resistance between the can coupling part **53** and the beading part **21** may decrease.

(76) A distance **R2** from the central axis of the electrode assembly **10** to the outer edge of the can coupling part **53** may be greater than the distance **R1** from the central axis of the electrode assembly **10** to the outer edge of the tab coupling part **52** in the radial direction of the battery can **20**. In addition, a distance **R3** from the central axis of the electrode assembly **10** to the inner edge of the can coupling part **53** may be less than the distance **R2** from the central axis of the electrode assembly **10** to the outer edge of the can coupling part **53** in the radial direction of the battery can **20**.

(77) In more detail, the outer edge of the can coupling part **53** may have an arc shape having a second curvature radius **R2** greater than the first curvature radius **R1** described above. Also, the inner edge of the can coupling part **53** may have an arc shape having a third curvature radius **R3** less than the second curvature radius **R2**. The center of curvature of the outer edge and the inner edge of the can coupling part **53** may substantially coincide with the central axis of the electrode assembly **10**.

(78) The second curvature radius **R2** may be greater than the inner radius **Rb** of the beading part **21** disposed on the battery can **20**, and the third curvature radius **R3** may be less than the inner radius **Rb** of the beading part **21**. That is, the distance **R2** from the central axis of the electrode assembly **10** to the outer edge of the can coupling part **53** may be greater than the inner radius **Rb** of the beading part **21**, and the distance **R3** from the central axis of the electrode assembly **10** to the inner edge of the can coupling part **53** may be less than the inner radius **Rb** of the beading part **21** in the radial direction of the battery can **20**.

(79) If the second curvature radius **R2** is equal to or less than the inner radius **Rb** of the beading part **21**, it may be difficult to couple the can coupling part **53** to the beading part **21**. If the third curvature radius **R3** is equal to or greater than the inner radius **Rb** of the beading part **21**, interference may occur between the bridge **54** to be described below and the beading part **21**.

(80) Both edges of the can coupling part **53** may be disposed along the radial direction of the first collector **50**. A virtual line extending along both the edges may pass through the center of the first collector **50**.

(81) The can coupling part **53** and the tab coupling part **52** may not overlap each other in the height direction of the battery can **20**. Thus, when a plurality of first collectors **50** are provided prior to one of them being coupled to the electrode assembly **10**, the can coupling parts **53** of one first collector **50** and the tab coupling part **52** of another first collector **50** may be prevented from interfering with each other.

(82) Coupling strength between the can coupling part **53** and the battery can **20**, more specifically, between the can coupling part **53** and the beading part **21**, may be greater than that between the tab coupling part **52** and the first electrode tab **11**. Thus, the can coupling part **53** may be stably coupled to the rigid battery can **20**.

(83) The bridge **54** may connect the can coupling part **53** to the center part **51**. The bridge **54** may extend radially outward from the center part **51** and may be spaced apart from the tab coupling part **52** in the circumferential direction of the first collector **50**.

(84) The bridge **54** may be provided in plurality, like the can coupling part **53**, and the plurality of bridges **54** may extend radially from the center part **51**. The bridges **54** and the tab coupling parts **52** may be alternately disposed in the circumferential direction of the first collector **50**.

(85) Since there is a height difference between the center part **51** and the can coupling part **53**, the bridge **54** may be inclined. The bridge **54** may obliquely extend in a direction away from the electrode assembly **10** as the distance from the center part **51** increases.

(86) An inclination of the bridge **54** may become smaller as the distance from the center part **51** increases. In more detail, the bridge **54** may include a first inclined part **54a** extending from the center part **51** and a second inclined part **54b** extending from the first inclined part **54a** and having a smaller inclination than the first inclined part **54a**.

(87) However, it is not limited thereto, and the bridge **54** may be rounded in a direction that becomes smaller as the distance from the center part **51** increases.

(88) Thus, it is possible to prevent the bridge **54** from being excessively deformed due to the sizing process and prevent the center part **51** and the first electrode tab **11** coupled to the center part **51** from being lifted. During the sizing process, since the beading part **21** is compressed toward the electrode assembly **10** to decrease in height, the tab coupling part **53** coupled to the beading part **21** may also move toward the electrode assembly **10**. As a result, repulsive force in a direction away from the electrode assembly **10** may act on the center part **51** connected to the tab coupling part **53** and the bridge **54**. If the inclination of the bridge **54** is constant or becomes steeper as it moves away from the center part **51**, the repulsive force may largely act on the center part **51**, and, thus, the center part **51** and the first electrode tab **11** coupled to the center part **51** may be lifted. On the other hand, the bridge **54** according to the present aspect may minimize the lifting of the center part **51** and the first electrode tab **11**, which is caused by the sizing process.

(89) In addition, a process of press-fitting the outer circumference of the battery can **20** inward to form the beading part **21** may be performed in a state in which the electrode assembly **10** to which the first collector **50** is coupled is accommodated in the battery can **20**. In this process, since the bridge **54** is decreasingly inclined as it moves away from the center part **51**, the beading part **21** may be easily formed without interfering with the bridge **54**.

(90) An edge of the cap plate **30** may be disposed between the beading part **21** and the crimping part **22** of the battery can **20** and may be fixed by a sealing part **72** to be described below.

(91) The cap plate **30** may not protrude beyond the battery can **20**. For example, when the cap plate **30** is disposed on a lower end of the battery can **20**, a bottom surface of the plate part **31** of the cap plate **30** may be disposed on the same surface as the bottom surface of the battery can **20** or be disposed higher than the bottom surface of the battery can **20**. Thus, the cap plate **30** may not be pressed upward by the bottom surface supporting the battery can **20**, and a phenomenon in which a pressure required for rupturing the venting part **34** due to a weight of the secondary battery **1** is different from a design value may be prevented from occurring.

(92) A venting part **34** may be provided in the cap plate **30**. The venting part **34** may have a thickness less than that of the surrounding area. Thus, the venting part **34** may be structurally weak compared to a surrounding area thereof, and if an internal pressure of the battery can **20** increases beyond a preset value, the venting part **34** may be preferentially ruptured.

(93) The venting part **34** may be disposed along a circumferential direction of the cap plate **30**. For example, the venting part **34** may have a closed loop shape such as a ring. Therefore, when the venting part **34** is ruptured, an area of the cap plate **30** disposed inside the venting part **34** may be easily separated to form an opening, and a gas in the battery can **20** may be quickly discharged.

(94) The venting part **34** may be formed by notching opposite surfaces of the cap plate **30** to a predetermined depth to partially reduce the thickness of the cap plate **30**. However, it is not limited thereto, and it may be possible to form the venting part **34** by notching only one surface of the cap plate **30**.

(95) In order to effectively absorb an impact or shaking applied to the secondary battery **1**, the

spacer **70** may have an elastic material.

(96) The spacer **70** may correspond to a distance between the first collector **50** and the cap plate **30**. In this case, the body part **71** may effectively prevent the electrode assembly **10** from moving within the battery can **20** due to a gap defined between the first collector **50** and the cap plate **30**. Therefore, the spacer **70** may prevent a coupled portion between the electrode assembly **10** and the first collector **50** and/or a coupled portion between the first collector **50** and the battery can **20** from being damaged.

(97) An opening facing a hollow C of the electrode assembly **10** may be defined in the spacer **70** through a center hole **51a** of the first collector **50**. The opening together with the center hole **51a** and the hollow C may function as a passage for inserting a welding rod or a passage for laser irradiation. In addition, the opening may function as a passage through which an electrolyte is smoothly impregnated into the electrode assembly **10** when the electrolyte is injected.

(98) The sealing part **72** may be integrated with the spacer **70**. However, this aspect is not limited thereto.

(99) The sealing part **72** may have a circular ring shape extending along an inner circumference of the battery can **20**. The sealing part **72** may be fixed between the beading part **21** and the crimping unit **22** of the battery can **20**. A portion of the sealing part **72** may be bent together with a crimping part **22** to fix an edge of the cap plate **30**. Thus, the sealing part **72** may firmly seal the gap between the cap plate **30** and the battery can **20**. As described above, the sealing part **72** may function as a gasket for improving fixing force of the cap plate **30** and sealing force of the battery can **20**.

(100) The sealing part **72** may press the can coupling part **53** of the first collector **50** toward the beading part **21** to further strengthen the coupling between the can coupling part **53** and the beading part **21**. In addition, the sealing part **72** may insulate the can coupling part **53** from the cap plate **30**.

(101) FIG. 5 is an enlarged cross-sectional view of a second collector and surroundings of the second collector, which are illustrated in FIG. 2, FIG. 6 is a plan view of the second collector according to an aspect of the present disclosure, and FIG. 7 is an enlarged view illustrating a narrow part illustrated in FIG. 6.

(102) As described above, the terminal **40** may pass through a central portion of the one surface **20a** of the battery can **20** and be coupled to the second collector **60** so as to be electrically connected to the second electrode tab **12**. In addition, the terminal **40** may be insulated from the battery can **20** by an insulating gasket G.

(103) In more detail, the terminal **40** may include a terminal part **41**, a body part **42** and a protrusion **43**.

(104) The terminal part **41** may be disposed outside the battery can **20**. The terminal part **41** may be a portion to which a bus bar disposed outside the battery can **20** can be connected. The terminal part **41** may have a substantially circular flat plate shape.

(105) The body part **42** may be integrated with the terminal part **41**. The body part **42** may pass through the battery can **20** and be coupled to the second collector **60**, which may include a tab coupling part **64** to be described in more detail below. The body part **42** may have a substantially circular columnar shape.

(106) The protrusion **43** may protrude radially outward from a circumference of the body part **42** and may be disposed inside the battery can **20**. The protrusion **43** may be integrated with the body part **42**. The protrusion **43** may be disposed to be inclined in a direction closer to the second collector **60** as it moves away from an outer circumference of the body part **42**.

(107) An end of the protrusion **43** may not protrude toward the second collector **60** more than an end of the body part **42**. Therefore, it is possible to prevent the second collector **60** from being damaged by the protrusion **43**.

(108) A diameter of each of the terminal part **41** and the protrusion **43** may be greater than a diameter of the body part **42**. As a result, the terminal **40** may be fixed so as not to be separated from the battery can **20**.

(109) The insulating gasket G may seal a gap between the terminal **40** and the battery can **20**. In more detail, the insulating gasket G may be bent along the terminal part **41** and the protrusion **43** and fixed between the battery can **20** and the terminal **40**.

(110) The insulator **80** may insulate the second collector **60** from the battery can **20**. The insulator **80** may be disposed between the second collector **60** and the battery can **20**.

(111) In more detail, the insulator **80** may include a plate part **81** and a circumferential part **82**.

(112) The plate unit **81** may be disposed between the second collector **60** and the one surface **20a** of the battery can **20**. The plate part **81** may have an approximately circular plate shape.

(113) A hole through which the terminal **40** passes may be defined in a center of the plate part **81**. The terminal **40** may be coupled to the second collector **60** toward the hole. An end of the protrusion **43** of the terminal **40** may face an inner circumference of the hole.

(114) The circumferential part **82** may protrude from an edge of the plate part **81** between the inner circumference of the battery can **20** and the outer circumference of the electrode assembly **10**. The circumferential part **82** may be integrated with the plate part **81**, but is not limited thereto.

(115) The circumferential part **82** may be disposed between the outer circumference of each of the second collector **60** and the second electrode tab **12** and the inner circumference of the battery can **20**. A height of the circumferential part **82** in the height direction of the battery can **20** may be greater than the sum of the heights of the second electrode tab **12** and the second collector **60**. Thus, the circumferential part **82** may reliably insulate the second electrode tab **12** and the second collector **60** from the battery can **20**.

(116) Central axes of the electrode assembly **10**, the battery can **20**, and the second collector **60** may substantially coincide with each other.

(117) The second collector **60** may be coupled to the second electrode tab **12**. In more detail, at least a portion of the second electrode tab **12** may be bent toward a hollow C of the electrode assembly **10**, and the second collector **60** may be coupled to the bent portion **12a** of the second electrode tab **12**. portions **12a** of the plurality of first electrode tabs **12** adjacent to each other may overlap each other.

(118) As a result, a height of the second electrode tab **12** may be reduced, and energy density of the secondary battery **1** may be improved. In addition, since a coupling area between the second electrode tab **12** and the second collector **60** increases, coupling force between the second electrode tab **12** and the second collector **60** may be improved to reduce resistance.

(119) In more detail, the second collector **60** may include an edge part **61**, a terminal coupling part **62** disposed inside the edge part **61** and coupled to the terminal **40**, a bridge **63** connecting the edge part **61** to the terminal coupling part **62**, and a tab coupling part **64** extending inward from the edge part **61**, spaced apart from the bridge **63** and the terminal coupling part **62**, and coupled to the second electrode tab **12**.

(120) The edge part **61** may have a substantially circular rim shape. However, this aspect is not limited thereto. The edge part **61** may be disposed between the second electrode tab **12** and the insulator **80** in the height direction of the battery can **20**.

(121) The terminal coupling part **62** may be disposed at an inner center of the edge part **61**. At least a portion of the terminal coupling part **62** may face the hollow C of the electrode assembly **10**.

(122) The terminal coupling part **62** may have a substantially circular shape. The terminal coupling part **62** may be coupled to the terminal **40**, more specifically, to the body part **42** of the terminal **40**. For example, the terminal coupling part **62** may be welded to a bottom surface of the body part **42** of the terminal **40**.

(123) The bridge **63** may connect the edge part **61** to the terminal coupling part **62**. The bridge **63** may extend radially inward from the edge part **61**.

(124) The bridge **63** may be provided in plurality, and the plurality of bridges **63** may extend radially from the terminal coupling part **62**.

(125) A width of an outer end **63a** of the bridge **63** may increase as it approaches the edge part **61**.

Thus, rigidity of a portion at which the bridge **63** and the circumferential part **61** are connected to each other may be improved. In addition, when the second collector **60** is transferred or transported, the outer end **63a** of the bridge **63** having relatively high rigidity may be gripped and transported stably.

(126) The tab coupling part **64** may extend radially inward from the edge part **61**. The tab coupling part **64** may be coupled to the second electrode tab **12**. For example, the tab coupling part **64** may be welded to the bent portion **12a** of the second electrode tab **12**.

(127) The tab coupling part **64** may be spaced apart from the terminal coupling part **62**. In more detail, the tab coupling part **64** may be spaced apart from the terminal coupling part **62** in the radial direction of the second collector **60**. In more detail, each of the tab coupling part **64** and the terminal coupling part **62** may be only connected to the circumferential part **61**, and the tab coupling part **64** and the terminal coupling part **62** may not be directly connected to each other. As a result, when an impact or vibration is applied to the secondary battery **1**, since stress applied to the coupling area between the tab coupling part **64** and the second electrode tab **12** and the coupled portion between the terminal coupling part **62** and the terminal **40** are dispersed, possibility of occurrence of damage of the second collector **60** may be minimized.

(128) The tab coupling part **64** may be spaced apart from the bridge **63**. In more detail, the tab coupling part **64** may be spaced apart from the bridge **63** in the circumferential direction of the second collector **60**.

(129) The tab coupling part **64** may be provided in plurality. In the case of the present aspect, the number of tab coupling parts **64** and the number of bridges **63** may be the same. The bridges **63** and the tab coupling parts **64** may be alternately disposed in the circumferential direction of the second collector **60**. Each of the bridges **63** may be disposed between a pair of tab coupling parts **64** adjacent to each other in the circumferential direction of the second collector **60**.

(130) A width w_2 of the bridge **63** may be less than or equal to a width w_4 of the tab coupling part **64**. As a result, electrical resistance may increase in the bridge **63**, and when overcurrent occurs, a portion of the bridge **63** is ruptured to block the overcurrent.

(131) In order to improve the overcurrent blocking function, a narrow part **63b** having a relatively narrow width may be provided in the bridge **63**. For example, the narrow part **63b** may be provided by recessing grooves in opposite edges of the bridge **63**.

(132) A width w_3 of the narrow part **63b** may be narrower than the width w_2 of the bridge **63**. Also, the width w_3 of the narrow part **63b** may be narrower than the width w_1 of the edge part **61**.

(133) Thus, when the electrical resistance increases in the narrow part **63b**, and the overcurrent flows through the bridge **63**, the narrow part **63b** is ruptured preferentially rather than the edge part **61** and the bridge **63** to reliably block the overcurrent. That is, the narrow part **63b** may perform a fuse function.

(134) In more detail, the width w_3 of the narrow part **63b** may be 70% to 80% of the width w_2 of the bridge **63**. Thus, the high reliability of the fuse function may be maintained while maintaining sufficient high rigidity of the narrow part **63b**. If the width w_3 of the narrow part **63b** is less than 70% of the width w_2 of the bridge **63**, there is a concern that the narrow part **63b** is easily ruptured when the impact or vibration is applied to the secondary battery **1**. In addition, if the width w_3 of the narrow part **63b** is greater than 80% of the width w_2 of the bridge **63**, there is a concern that the narrow part **63b** may not ruptured preferentially when the overcurrent flows.

(135) The narrow part **63b** may face the bent portion **12a** of the second electrode tab **12** with respect to the height direction of the electrode assembly **10**. Thus, the bent portion **12a** of the second electrode tab **12** may prevent residues generated when the narrow part **63b** is ruptured from being introduced into the electrode assembly **10**.

(136) The narrow part **63b** may be disposed closer to the outer end **63a** of the ends of the bridge **63**.

(137) A distance d from the center of the second collector **60** to the narrow part **63b** is 0.4 times to 0.9 times the radius R_a of the electrode assembly **10** in the radial direction of the second collector

60. That is, a distance d from the central axis of the electrode assembly **10** to the narrow part **63b** is 0.4 times to 0.9 times the radius R_a of the electrode assembly **10** in the radial direction of the electrode assembly **10**.

(138) As described above, the bent portions **12a** of the second electrode tab **12** may overlap each other. In this case, in some areas adjacent to the outer circumference of the electrode assembly **10** and in some areas adjacent to the hollow **C**, an overlapping degree or number of bent portions **12a** of the second electrode tab **12** may be reduced. For this reason, if the distance d from the center of the second collector **60** to the narrow part **63b** is less than 0.4 times or larger than 0.9 times the radius R_a of the electrode assembly **10**, the residues generated when the narrow part **63b** is ruptured may be introduced into the electrode assembly **10**.

(139) On the other hand, as in the present aspect, when the distance d from the center of the second collector **60** to the narrow part **63b** is 0.4 times to 0.9 times the radius R_a of the electrode assembly **10**, the narrow part **63b** may face the area on which the bent portions **12a** of the second electrode tab **12** sufficiently overlap each other in the height direction of the electrode assembly **10**. Thus, it is possible to reliably prevent the residues generated when the narrow part **63b** is ruptured from being introduced into the electrode assembly **10**.

(140) At least one opening **65** may be defined between the edge part **61** and the terminal coupling part **62**. When there are a plurality of bridges **63**, a plurality of openings **65** divided into a plurality of areas by the plurality of bridges **63** may be defined. The tab coupling part **64** may reduce a surface area of the opening **65**.

(141) In more detail, a ratio occupied by a surface area of the at least one opening **65** to the total surface area of the second collector **60** may be 40% to 99%. The total surface area of the second collector **60** may mean that sum of all values of respective surface areas of the edge part **61**, the terminal coupling part **62**, the bridge **63**, the tab coupling part **64**, and the opening portion **65**. Thus, impregnation of the electrolyte into the electrode assembly **10** may be improved by the opening **65**. If the surface area of at least one opening **65** is less than 40% of the total surface area of the second collector **60**, the impregnation of the electrolyte into the electrode assembly **10** may be deteriorated.

(142) The inner end **64a** of the tab coupling part **64** may be spaced a predetermined apart from the terminal coupling part **62** to define a predetermined gap. A width of the inner end **64a** of the tab coupling part **64** may decrease as it approaches the terminal coupling part **62**. As a result, the surface area of the opening **65** may further increase, and the impregnation of the electrolyte into the electrode assembly **10** may be improved.

(143) A width w_1 of the edge part **61** may be narrower than a width w_4 of the tab coupling part **64**. Thus, it is possible to secure a sufficiently wide coupling area between the tab coupling part **64** and the second electrode tab **12** while maintaining the sufficient wide area of the open portion **65**.

(144) A radius R_5 of the second collector **60** may be 33% to 102% of the radius R_a of the electrode assembly **10**. In addition, the radius R_5 of the second collector **60** may be 33% to 98.5% of an inner radius R_d of the battery can **20**. The radius R_5 of the second collector **60** may mean a distance from the center of the second collector **60** to the outer circumference of the edge part **61**.

(145) If the radius R_5 of the second collector **60** is less than 33% of the radius R_a of the electrode assembly **10** or less than 33% of the inner radius R_d of the battery can **20**, a function as an electrical passage of the second collector **60** connecting the second electrode tab **12** to the terminal **40** may be deteriorated.

(146) On the other hand, when the radius R_5 of the second collector **60** is greater than 102% of the radius R_a of the electrode assembly **10** or greater than 98.5% of the inner radius R_d of the battery can **20**, the edge part **61** of the second collector plate **60** may interfere with the edge part **82** of the insulator **80**, and if a size of the battery can **20** increases, or a size of the electrode assembly **10** decreases in order to prevent the interference, the energy density of the secondary battery **1** may decrease.

(147) The radius R_4 of the terminal coupling part **62** may be 40% to 320% of the radius R_c of the

body part **42** coupled to the terminal coupling part **62** at the terminal **40**. In more detail, the radius **R4** of the terminal coupling part **62** may be 40% to 320% of the radius **Rc** of the bottom surface of the body part **42**.

(148) If the radius **R4** of the terminal coupling part **62** is less than 40% of the radius **Rc** of the body part **42**, the coupling area between the terminal coupling part **62** and the terminal **40** becomes too small, resulting in increasing in resistance. On the other hand, when the radius **R4** of the terminal coupling part **62** is greater than 320% of the radius **Rc** of the body part **42**, a length of the tab coupling part **64** spaced apart from the terminal coupling part **62** may be shortened, and the surface area of the opening **65** may be reduced. As a result, the coupling area between the second electrode tab **12** and the tab coupling part **64** may decrease, which could lead to an increase in resistance, and the impregnation of the electrolyte into the electrode assembly **10** may decrease.

(149) A difference between the coupling area between the first electrode tab **11** and the first collector **50** (see FIG. 3) and the coupling area between the second electrode tab **12** and the second collector **60** may be three times or less. For convenience of explanation, the coupling area between the first electrode tab **11** and the first collector **50** may be referred as a first coupling area, and the coupling area between the second electrode tab **12** and the second collector **60** may be referred to as a second coupling area.

(150) The first coupling area may be a coupling area between the first electrode tab **11** and the tab coupling part **52** (see FIG. 4) of the first collector **50**. The second coupling area may be a coupling area between the second electrode tab **12** and the tab coupling part **64** of the second collector **60**.

(151) Any one of the first coupling area and the second coupling area may be 3 times or less of the other one. Preferably, the first coupling area may be the same as or similar to the second coupling area. Thus, excessive increase in resistance of either the first collector **50** or the second collector **60** may be prevented from occurring.

(152) The second collector **60** may not overlap the beading part **21** (see FIG. 3) in the height direction of the electrode assembly **10**.

(153) Alternatively, the second collector **60** may overlap the beading part **21** in the height direction of the electrode assembly **10**, and a width of an area overlapping the second collector **60** on the beading part **21** may be 80% or less of a width **wb** of the beading part **21**. That is, the second collector **60** may overlap by 80% or less of the width **wb** of the beading part **21** in the height direction of the electrode assembly **10**. In this case, the width **wb** of the beading part **21** may mean the width of the battery can **20** in the radial direction.

(154) If the second collector **60** overlaps by more than 80% of the width **wb** of the beading part **21** in the height direction of the electrode assembly **10**, the second collector **60** may interfere with the circumferential part **82** of the insulator **80**, and thus, in order to prevent the interference, if the battery can increases in size, or the electrode assembly **10** decreases in size, the energy density of the secondary battery **1** may be reduced.

(155) FIG. 8 is a view illustrating a modified example of the narrow part illustrated in FIG. 6.

(156) As described above, the narrow part **63b** may be provided by recessing the grooves on opposite edges of the bridge **63**. However, it is not limited thereto, and as illustrated in FIG. 8, the narrow part **63b** may be provided by providing a hole in the bridge **63**.

(157) In more detail, a hole may be defined in the central portion of the bridge **63** in the width direction, and opposite sides of the hole may constitute the narrow part **63b**. In this case, the width **w3** of the narrow part **63b** may mean a sum of a width **w31** of one side of the hole and a width **w32** of the other side of the hole. Regarding the width **w3** of the narrow part **63b**, the above description is cited.

(158) FIG. 9 is a view illustrating a state in which a cover member surrounds the narrow part illustrated in FIG. 6.

(159) A cover member **63c** made of an insulating material may be provided on the bridge **63** of the second collector **60** to cover the narrow part **63b**. For example, the cover member **63c** may be a

tape taped to cover the narrow part **63b**.

(160) The cover member **63c** may prevent the residues generated when the bridge **63**, particularly the narrow part **63b** is ruptured by the overcurrent from being spread to the surroundings.

(161) In addition, since the cover member **63c** deteriorates heat dissipation of the narrow part **63b**, if the overcurrent flows through the bridge **63**, an amount of heat generated in the narrow part **63b** may further increase, and the narrow part **63b** may be ruptured more quickly. In order to effectively deteriorate the heat dissipation of the narrow part **63b**, thermal conductivity of the cover member **63c** may be less than that of the bridge **63**.

(162) For example, the cover member **63c** may include a polyimide (PI) material. Since the polyimide (PI) material has a melting point of approximately 375 degrees Celsius to 401 degrees Celsius, the cover member **63c** may not be melted due to the heat generated by the bridge **63**.

(163) Unlike the structure illustrated in FIG. **9**, a structure in which the narrow part **63b** is not provided in the bridge **63**, and the cover member **63c** covers a specific point of the bridge **63** is also possible. In this case, since the cover member **63c** deteriorates the heat dissipation at a specific point of the bridge **63**, when the overcurrent flows through the bridge **63**, an amount of heat generated at the specific point may increase and thus be preferentially ruptured. That is, the specific point of the bridge **63** surrounded by the cover member **63c** may perform a fuse function.

(164) Therefore, a portion of the previous description of the narrow part **63b** may be cited for the cover member **63c**. More specifically, the cover member **63c** may be disposed closer to the outer end **63a** of the ends of the bridge **63**. Also, the cover member **63c** may face the bent portion **12a** of the second electrode tab **12** in the axial direction of the electrode assembly **10**. In addition, a distance *d* from the center of the second collector **60** to the cover member **63b** is 0.4 times to 0.9 times the radius *R_a* of the electrode assembly **10** in the radial direction of the second collector **60**.

(165) FIG. **10** is a plan view of a second collector according to another aspect of the present disclosure.

(166) Hereinafter, contents duplicated with the previously described contents will be cited, and described with a focus on the differences.

(167) In a second collector **60** according to the present aspect, the number of bridges **63** may be less than the number of tab coupling parts **64**.

(168) In more detail, some of the plurality of tab coupling parts **64** may be continuously disposed in a circumferential direction of the second collector **60**. That is, the tab coupling parts **64** and the bridges **63** may not be alternately disposed in the circumferential direction of the second collector **60**. The bridge **63** may be disposed between a pair of tab coupling parts **64** adjacent to each other in the circumferential direction of the second collector **60**.

(169) Thus, since an amount of current flowing through each bridge **63** increases, sensitivity of a narrow part **63b** to overcurrent may increase. Therefore, the narrow part **63b** may be ruptured more quickly, and safety of the secondary battery **1** may be improved.

(170) The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other aspects, which fall within the true spirit and scope of the present disclosure.

(171) Thus, the aspect of the present disclosure is to be considered illustrative, and not restrictive, and the technical spirit of the present disclosure is not limited to the foregoing aspect.

(172) Therefore, the scope of the present disclosure is defined not by the detailed description of the disclosure but by the appended claims, and all differences within the scope will be construed as being included in the present disclosure.

(173) TABLE-US-00001 [Description of the Symbols] 1: Cylindrical secondary battery 10:

Electrode assembly 11: First electrode tab 12: Second electrode tab 20: Battery can 21: Beading

part 22: Crimping part 30: Cap plate 34: Venting part 40: Terminal 41: Terminal part 42: Body part

43: Protrusion 50: First collector 51: Center part 51a: Center hole 52: Tab coupling par 52d:

Injection hole 53: Can coupling part 54: Bridge 60: Second collector 61: Edge part 62: Terminal

coupling part 63: Bridge 63a: Outer end (of bridge) 63b: Narrow part 63c: Cover member 64: Tab coupling part 64a: Inner end (of tab coupling part) 65: Opening 70: Spacer 72: Sealing part 80: Insulator 81: Plate part 82: Circumferential part

Claims

1. A cylindrical secondary battery comprising: an electrode assembly having an electrode tab; a battery can configured to accommodate the electrode assembly therein; a terminal configured to pass through the battery can while being spaced from the battery can; and a collector configured to electrically connect the electrode tab to the terminal, the collector including: an edge part; a terminal coupling part located inside the edge part, the terminal coupling part being coupled to the terminal; a bridge connecting the edge part to the terminal coupling part; and a tab coupling part extending inward from the edge part, the tab coupling part being spaced apart from the bridge and the terminal coupling part, the tab coupling part being coupled to the electrode tab, wherein the bridge includes a narrow part located between opposite ends of the bridge, wherein at least a portion of the electrode tab is bent toward a central axis of the electrode assembly, and wherein the narrow part is opposite the bent portion of the electrode tab in an axial direction of the electrode assembly.
2. The cylindrical secondary battery of claim 1, wherein the bridge is provided in plurality, wherein the tab coupling part is provided in plurality, and wherein the bridges and the tab coupling parts are alternately disposed in a circumferential direction of the collector.
3. The cylindrical secondary battery of claim 1, wherein an outer end of the bridge has a width that gradually increases as it approaches the edge part.
4. The cylindrical secondary battery of claim 1, wherein the narrow part has a width corresponding to 70% to 80% of a width of the bridge at one of the ends of the bridge.
5. The cylindrical secondary battery of claim 1, wherein the narrow part is disposed closer to an outer end of the opposite ends of the bridge.
6. The cylindrical secondary battery of claim 1, wherein a distance from a center of the collector to the narrow part is 0.4 times to 0.9 times a radius of the electrode assembly.
7. The cylindrical secondary battery of claim 1, wherein the narrow part is defined by recessed grooves at opposite edges of the bridge or as a hole extending through the bridge.
8. The cylindrical secondary battery of claim 1, wherein the narrow part has a width less than a width of the bridge between the edges of the bridge.
9. The cylindrical secondary battery of claim 1, further comprising a cover member configured to surround the bridge, the cover member being made of an insulating material having thermal conductivity less than that of the bridge.
10. The cylindrical secondary battery of claim 9, wherein the cover member is made of a polyimide (PI) material.
11. The cylindrical secondary battery of claim 1, wherein the edge part has a width less than a width of the tab coupling part.
12. The cylindrical secondary battery of claim 1, wherein the collector has a radius corresponding to 33% to 102% of a radius of the electrode assembly.
13. The cylindrical secondary battery of claim 1, wherein the collector has a radius corresponding to 33% to 98.5% of an inner radius of the battery can.
14. The cylindrical secondary battery of claim 1, wherein the terminal includes a body part coupled to the terminal coupling part, and wherein the terminal coupling part has a radius corresponding to 40% to 320% of a radius of the body part.
15. The cylindrical secondary battery of claim 1, wherein at least one opening is defined between the edge part and the terminal coupling part, and wherein a ratio of a surface area of the at least one opening to a total surface area of the collector is 40% to 99%.

16. The cylindrical secondary battery of claim 1, wherein an inner end of the tab coupling part has a width that gradually decreases as it approaches the terminal coupling part.
17. A collector configured to electrically connect an electrode assembly to a terminal passing through a battery can in which the electrode assembly is accommodated, the collector comprising: an edge part; a terminal coupling part located inside the edge part, the terminal coupling part being coupled to the terminal; a bridge connecting the edge part to the terminal coupling part; and a tab coupling part extending inward from the edge part, the tab coupling part being spaced apart from the bridge and the terminal coupling part, the tab coupling part being configured to be coupled to an electrode tab provided in the electrode assembly, wherein the bridge includes a narrow part located between opposite ends of the bridge wherein at least a portion of the electrode tab is bent toward a central axis of the electrode assembly, and wherein the narrow part is opposite the bent portion of the electrode tab in an axial direction of the electrode assembly.
18. The collector of claim 17, wherein the bridge is provided in plurality, wherein the tab coupling part is provided in plurality, and wherein the bridges and the tab coupling parts are alternately disposed in a circumferential direction of the collector.
19. The collector of claim 17, wherein an outer end of the bridge has a width that gradually increases as it approaches the edge part.
20. The collector of claim 17, wherein the narrow part is defined by recessed grooves at opposite edges of the bridge or as a hole extending through the bridge.
21. The collector of claim 17, further comprising a cover member configured to surround the bridge, the cover member being made of an insulating material having thermal conductivity less than that of the bridge.
22. The collector of claim 17, wherein an inner end of the tab coupling part has a width that gradually decreases as it approaches the terminal coupling part.
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